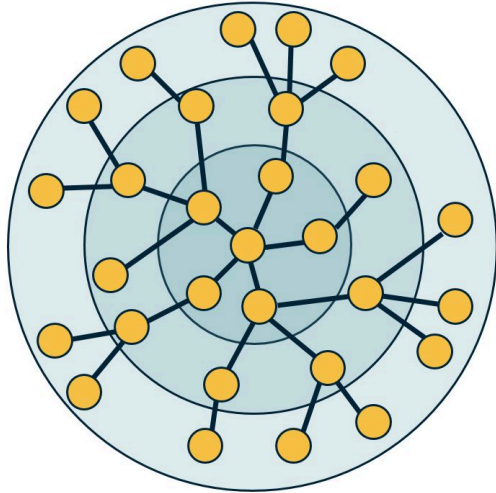


L5: Diameter and CPL of G(n,p) Networks



Let us now go back to the simplest random graph model we have studied so far, i.e., the ER or G(n,p) model. What is the diameter of such a network?

We can derive an approximate expression as follows.

Suppose that the average degree is $\bar{k} = (n-1)p > 1$ (so that the network has a giant connected component). We will further assume that the topology of the network is a **tree**.

We start from node i . Within one hop away from that node, we expect to visit $\bar{k}$ nodes. Within two hops, we expect to visit approximately $(\bar{k})^2$ nodes. And with a similar reasoning, after s hops we expect to visit approximately $(\bar{k})^s$ nodes.

The total number of nodes in the network is n however, and we expect to visit all of them with the maximum number of hops that is possible, which is the network diameter D .

So, $n \approx (\bar{k})^D$. Solving for D , we get that

$$D \approx \frac{\ln n}{\ln \bar{k}}$$

Even though this expression is a very rough approximation, it shows something remarkable: in a random network, even the longest shortest—paths are expected to grow very slowly (i.e., *logarithmically*) with the size of the network.

Here is a numerical example: suppose we have a social network that covers all people (around 7 billion today) and assumes that each person knows well 64 other people. According to the previous expression, any two people are connected to each other with a “social chain” (shortest-path) of 5.4 relations.

Additionally, the CPL is lower (or equal) than the diameter of the network. So the average shortest-path length is also upper bounded by the same logarithmic expression we derived above.

The diameter of G(n,p) networks – more accurate expressions

Two approximate expressions that are often used in practice apply for sparse and dense networks, respectively.

Specifically, for very sparse networks, as k approaches to 1 from above (so that the network is still expected to have a large connected component), the diameter is expected to be

$$D \approx 3 \frac{\ln n}{\ln \bar{k}}$$

This is three times larger than the expression we derived in the previous page. For very dense networks, on the other hand, a good approximation for the diameter is:

$$D \approx \frac{\ln n}{\ln \bar{k}} + \frac{2 \ln n}{\bar{k}} + \ln n \frac{\ln \bar{k}}{(\bar{k})^2}$$

Note that in both cases, the diameter is still increasing with the logarithm of the network size. So, the main qualitative conclusion remains what we stated in the previous page, i.e., the diameter of G(n,p) networks increases very slowly (*logarithmically*) with the number of nodes – and so the CPL cannot increase faster than that either.

Food For Thought

- Enumerate all the assumptions we made in the derivation for the diameter of G(n,p) networks.

- Go back to the example of the previous page (*a social network with 7 billion nodes and an average degree of 64*). What would be the diameter according to either of the previous two approximations?